Isaac ROS Depth Segmentation

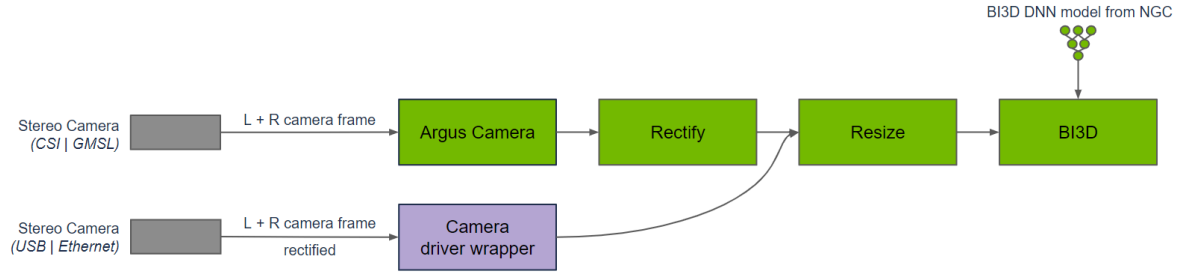
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Isaac ROS Depth Segmentation official website link: https://nvidia-isaac-ros.github.io/repositories_and_packages/isaac_ros_depth_segmentation/index.html

Overview

Isaac ROS Depth Segmentation provides NVIDIA-accelerated packages for depth segmentation. Bi3D uses the optimized DNN model to perform binarized depth estimation on stereo images, which can determine whether an obstacle is within a proximity field and estimate the ground free space.



Quick Experience

First use the command below to start the container

```
./run_isaac.sh
```

Then confirm the container on the host with `docker ps`, and open four container terminals with `docker exec -it <container name> bash`; each terminal should use the same ROS/FastDDS environment, for example:

```
Jetson Orin Nano L4T R39.2.1 · (8GB + 8GB ZRAM)
RDP 192.168.2.61:3389 jetson/yahboom [LAN]
VNC 192.168.2.61:5900 pass: yahboom [LAN]
jetson@jetson:~$ docker ps
CONTAINER ID   IMAGE                          COMMAND                  CREATED
      STATUS          PORTS     NAMES
8f397e4b6c3e   my-isaac-ros-slam:4.6-all   "/bin/bash -c '\ncat …"   42 seconds
ago   Up 41 seconds             magical_bhabha
jetson@jetson:~$
```

Note: The terminals below all execute inside the container:

> After the power mode is set to MAXN, an overcurrent warning may pop up when running the case; ignore it.

Terminal A: Start Nano pool=8 ESS. The input image is 1920×1080, and the model input is 960×576:

```
ros2 launch /workspaces/isaac_ros-dev/launch/ess_nano_pool8.launch.py \
  engine_file_path:=/workspaces/isaac_ros-
dev/isaac_ros_assets/models_pending/dnn_stereo_disparity/dnn_stereo_disparity_v4.
1.0_onnx_trt10.16/ess.engine \
  ess_plugin_file_path:=/workspaces/isaac_ros-
dev/isaac_ros_assets/models_pending/dnn_stereo_disparity/dnn_stereo_disparity_v4.
1.0_onnx_trt10.16/plugins/aarch64/ess_plugins.so \
  input_image_width:=1920 input_image_height:=1080 \
  model_input_width:=960 model_input_height:=576
```

Terminal B: Play the input in a loop at 0.25× speed:

```
ros2 bag play /workspaces/isaac_ros-
dev/isaac_ros_assets/isaac_ros_ess/rosbags/ess_rosbag \
  --loop --rate 0.25 --remap \
  /left/camera_info:=/left/camera_info_rect \
  /right/camera_info:=/right/camera_info_rect
```

Terminal C: Run the depth segmentation post-processing:

```
python3 /workspaces/isaac_ros-dev/ess_segmentation.py
```

Terminal D: Open rqt image view and select `/ess/depth_segmentation` in Topic:

```
ros2 run rqt_image_view rqt_image_view
```

